

Data Analytics with Python Certification Training

Course-End Project Problem Statement

Course-End Project - 2

Marketing Campaigns

Overview

In this exercise, the focus is on understanding the marketing mix and its impact on customer acquisition. The data provided contains information on various aspects of the four Ps of marketing: product, price, place, and promotion. By conducting exploratory data analysis (EDA) and hypothesis testing, you aim to gain insights into the factors influencing customer acquisition.

Instructions

- Review the learning materials in Data Analytics with Python
- Carefully read the situation, tasks, actions, and result sections to grasp the assessment fully
- Complete and submit your assessment via the Learning Management System (LMS)
- Follow the provided guidelines closely, ensuring your report includes all required analyses and interpretations

Situation

As a data analyst, you analyze marketing data to better understand customer acquisition. This involves examining variables such as birth year, education, income, expenditures on various products, sales channels, and promotional outcomes. By delving into these aspects, you can uncover patterns and trends that will guide marketing strategies to improve customer acquisition.

Task

As a data analyst, your task involves conducting exploratory data analysis and hypothesis testing to understand the impact of various marketing factors (product, price, place, and promotion) on customer acquisition.

Actions

1. Data Import

- Import the necessary libraries (**pandas**, **matplotlib.pyplot**, **seaborn**, **numpy**, and **scipy.stats**)
- Read the CSV file (**marketing_data.csv**) using **pd.read_csv()** and store it in a DataFrame (**data**)
- Print a snapshot of the data (**data.head()**) and check the data types and missing values (**data.info()**)

2. Investigate Variables

- Check variables like **Dt_Customer** and **Income** to ensure they are imported correctly
- Convert the **income** column to a float by removing '\$' and ',' and using **astype('float')**
- Convert the **Dt_Customer** column to a datetime object using **pd.to_datetime()**

3. Missing Value Imputation

- Check for missing values using **data.isnull().sum()**
- Clean the **marital_status** and **education** categories (e.g., combining similar categories)
- Impute missing **income** values based on similar **education** and **marital_status** categories' average incomes

4. Create Variables

- Calculate the **age** of customers based on **year_birth**
- Calculate **spending** as the sum of spending columns (**mntwines**, **mntfruits**, etc.)
- Calculate **children_count** as the sum of **kidhome** and **teenhome**.
- Create a binary variable **has_children** indicating if a customer has children
- Derive **total_purchases** from the sum of purchase columns

5. Box Plots and Histograms

- Plot box plots and histograms for numerical variables (**income**, **spending**, **age**, **total_purchases**) to understand their distributions and identify outliers
- Perform outlier treatment by capping the outliers to the upper and lower limits

6. Encoding

- Use ordinal encoding for the **education** column
- Use one-hot encoding for the **country** and **marital_status** columns

7. Correlation Analysis

- Select a subset of columns for correlation analysis
- Compute the correlation matrix (**new_data[var_cor].corr()**) and plot a heatmap (**sns.heatmap()**)

8. Hypothesis Testing

- Define and test hypotheses related to customer behavior using statistical tests (**stats.ttest_ind()**)

9. Visualization

- Visualize the performance of products (**prod_data**) using a bar plot
- Explore the relationship between **age** and campaign acceptance rate (**acceptedcmp5**) using a bar plot
- Determine the country with the most customers who accepted the last campaign using a bar plot
- Analyze the relationship between **children_count** and **spending** using a strip plot
- Visualize the educational background of customers who complained in the last two years using a bar plot
- Explore the relationship between **income** and **spending** using a scatter plot

Result

Build a data-driven report that uncovers valuable insights from the marketing data. These insights can help optimize marketing strategies, refine product offerings, and enhance customer engagement. This data-driven approach enables businesses to make informed decisions and drive growth in customer acquisition.